

Majorly Used Agricultural Sprinklers and Their Problems

Most Common Agricultural Sprinkler Problems

Sprinkler Type	Common Usage	Major Problems	Main Causes
Impact Sprinkler	Field crops, large farms	Hammer arm sticking, irregular rotation, nozzle wear	Rust, spring wear, sand particles
Rotary Sprinkler	Agricultural fields	Gear wear, rotation failure, uneven rotation	Debris, mechanical wear
Rotor Sprinkler	Large crop areas	Arc adjustment failure, seal leakage	Pressure variation, internal gear damage
Center Pivot Sprinkler	Large circular farms	Tower misalignment, wheel motor failure, pressure drop	Electrical faults, motor issues
Lateral Move Sprinkler	Rectangular farms	Wheel synchronization failure, pipe stress	Motor mismatch, alignment issues
Traveling Sprinkler	Large farms	Wheel slipping, hose tangling, gearbox failure	Mechanical wear, improper terrain
Big Gun Sprinkler	Sugarcane, fodder crops	High pressure requirement, nozzle clogging, recoil issues	Pressure fluctuation, dirty water

Rain Gun Sprinkler	Large open fields	Uneven spray in wind, nozzle wear	Wind drift, abrasive particles
Micro Sprinkler	Orchards, nurseries	Tiny nozzle blockage, low-pressure sensitivity	Fine particles, pressure drop
Mini Sprinkler	Vegetable cultivation	Uneven droplet distribution, clogging	Sediments, mineral deposits
Drip Sprinkler / Drip Irrigation	Orchards, row crops	Emitter clogging, algae formation, salt buildup	Dirty water, fertilizer residue
Pop-Up Sprinkler	Turf farms, landscaped agriculture	Head not popping up, spring failure	Dirt accumulation, damaged spring
Fixed Spray Sprinkler	Small farms, gardens	Uneven spray, mist formation	Pressure imbalance, clogged nozzle
Oscillating Sprinkler	Small agricultural plots	Oscillation stop, hole blockage	Rust, dirt buildup
Portable Sprinkler	Temporary irrigation	Hose leakage, poor stability	Manual handling damage
Underground Sprinkler	Permanent farm irrigation	Hidden leakage, root intrusion	Soil movement, tree roots

Problem	Commonly Seen In
Nozzle clogging	Impact, rain gun, micro, drip
Pressure variation	Rotor, rain gun, center pivot
Rotation failure	Rotary, impact, rotor

Wheel/motor failure	Center pivot, lateral move
Pipe leakage	Underground, portable
Wind drift	Rain gun, big gun
Gear wear	Rotary, traveling sprinkler
Sediment blockage	Drip, mini sprinkler
Alignment issue	Center pivot, lateral move
Spring failure	Pop-up sprinkler

Problem / Material	Why Filters Cannot Remove	Why Acid Cannot Remove	Common in Agriculture?
Dissolved salts	Salts are fully dissolved in water	Acid does not remove dissolved salts	Very common in borewell water
Biofilm (bacterial slime)	Sticks to pipe walls and emitters	Acid weak against biological slime	Very common in drip irrigation
Algae growth inside pipeline	Small fragments pass through	Acid does not stop regrowth well	Common in open tanks/ponds
Silica deposits	Very fine and hard	Most acids cannot dissolve silica	Common in groundwater
Fine clay / colloidal particles	Too small for normal filters	Acid cannot dissolve clay	Common in river/canal water
Iron bacteria slime	Sticky biological iron deposits	Acid only partially effective	Common in iron-rich borewells

Organic matter / decayed plant material	Sticky and soft, may bypass filters	Acid ineffective	Common in farm ponds
Root intrusion	Physical root growth	Acid ineffective	Happens in buried pipelines
Hard bio-mineral mixed clogging	Combination of slime + minerals	Acid removes only mineral part	Major emitter clogging issue
Pipe wall corrosion damage	Structural issue	Cannot be repaired chemically	Older GI pipelines